

hw3

September 12, 2026

1 Biomedical Data Science - Assignment 3

Shane Ferrell

```
[1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
sns.set()
```

```
[2]: df = pd.read_csv("kirby21AllLevels.csv")
```

```
[3]: df
```

```
[3]:      Unnamed: 0      rawid      roi  volume  min  max  \
0              1  kirby127a_3_1_ax.img  Telencephalon_L  531111  0.0  374.0
1              2  kirby127a_3_1_ax.img  Telencephalon_R  543404  0.0  300.0
2              3  kirby127a_3_1_ax.img  Diencephalon_L    9683  15.0  295.0
3              4  kirby127a_3_1_ax.img  Diencephalon_R    9678  10.0  335.0
4              5  kirby127a_3_1_ax.img   Mesencephalon   10268  55.0  307.0
...          ...          ...          ...          ...          ...
16715      16716  kirby959a_3_1_ax.img  Caudate_tail_L     406  65.0  248.0
16716      16717  kirby959a_3_1_ax.img  Caudate_tail_R     311  70.0  270.0
16717      16718  kirby959a_3_1_ax.img  Chroid_LVetc_L     399  15.0  229.0
16718      16719  kirby959a_3_1_ax.img  Chroid_LVetc_R     323  14.0  211.0
16719      16720  kirby959a_3_1_ax.img    IV_ventricle    3107   1.0  249.0
```

```
      mean    std  type  level  id    icv    tbv
0    128.3013  51.8593    1     1  127  1378295  1268519
1    135.0683  53.6471    1     1  127  1378295  1268519
2    193.5488  32.2733    1     1  127  1378295  1268519
3    193.7051  32.7869    1     1  127  1378295  1268519
4    230.8583  29.2249    1     1  127  1378295  1268519
...      ...      ...      ...      ...      ...
16715  167.1302  34.9645    2     5  959  1313114  1207218
16716  175.3475  39.5329    2     5  959  1313114  1207218
16717  120.6084  35.9975    2     5  959  1313114  1207218
16718  111.9703  31.4684    2     5  959  1313114  1207218
16719   70.7860  44.2799    2     5  959  1313114  1207218
```

[16720 rows x 13 columns]

```
[4]: kirby906a_ax_df = df.loc[df['rawid'] == 'kirby906a_ax.img'].copy()
kirby906a_ax_df
```

```
[4]:
```

	Unnamed: 0	rawid	roi	volume	min	max	\
12540	12541	kirby906a_ax.img	Telencephalon_L	467063	2.0	350.0	
12541	12542	kirby906a_ax.img	Telencephalon_R	470488	2.0	337.0	
12542	12543	kirby906a_ax.img	Diencephalon_L	8801	60.0	327.0	
12543	12544	kirby906a_ax.img	Diencephalon_R	9054	63.0	415.0	
12544	12545	kirby906a_ax.img	Mesencephalon	9564	86.0	352.0	
...	...	...	...	...	...	...	
13371	13372	kirby906a_ax.img	Caudate_tail_L	424	112.0	279.0	
13372	13373	kirby906a_ax.img	Caudate_tail_R	386	83.0	286.0	
13373	13374	kirby906a_ax.img	Chroid_LVetc_L	101	56.0	255.0	
13374	13375	kirby906a_ax.img	Chroid_LVetc_R	84	53.0	271.0	
13375	13376	kirby906a_ax.img	IV_ventricle	1835	3.0	275.0	

	mean	std	type	level	id	icv	tbv
12540	165.2599	57.1707	1	1	906	1195015	1123076
12541	171.8695	59.3001	1	1	906	1195015	1123076
12542	227.1878	31.2303	1	1	906	1195015	1123076
12543	231.6770	31.1780	1	1	906	1195015	1123076
12544	269.1003	28.6454	1	1	906	1195015	1123076
...	...	...	...	...	...	...	...
13371	182.8215	31.9975	2	5	906	1195015	1123076
13372	186.3707	37.6639	2	5	906	1195015	1123076
13373	181.6190	39.8132	2	5	906	1195015	1123076
13374	181.9857	43.3901	2	5	906	1195015	1123076
13375	108.3120	53.6962	2	5	906	1195015	1123076

[836 rows x 13 columns]

```
[5]: telencephalon_L_volume = kirby906a_ax_df.loc[
      (kirby906a_ax_df['roi'] == 'Telencephalon_L') &
      (kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 1)
    ]['volume'].sum()
print('type I level I telencephalon left volume: ', telencephalon_L_volume)

telencephalon_R_volume = kirby906a_ax_df.loc[
      (kirby906a_ax_df['roi'] == 'Telencephalon_R') &
      (kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 1)
    ]['volume'].sum()
print('type I level I telencephalon right volume: ', telencephalon_R_volume)
```

```
type I level I telencephalon left volume: 467063
type I level I telencephalon right volume: 470488
```

```
[6]: telencephalon_volume_total = kirby906a_ax_df.loc[
    (kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 1) &
    ((kirby906a_ax_df['roi'] == 'Telencephalon_L') | (kirby906a_ax_df['roi'] == 'Telencephalon_R'))
][['volume']].sum()

ICV_total = kirby906a_ax_df.loc[
    (kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 1)
][['volume']].sum()

TBV_total = kirby906a_ax_df.loc[
    (kirby906a_ax_df['type'] == 1) & (kirby906a_ax_df['level'] == 1) &
    (kirby906a_ax_df['roi'] != 'CSF')
][['volume']].sum()

ICV_fraction = telencephalon_volume_total / ICV_total
TBV_fraction = telencephalon_volume_total / TBV_total

print("type I level I telencephalon ICV:", ICV_fraction)
print("type I level I telencephalon TBV:", TBV_fraction)
```

type I level I telencephalon ICV: 0.7845516583473847

type I level I telencephalon TBV: 0.8348063710737297

```
[7]: all_regions_df = kirby906a_ax_df.loc[kirby906a_ax_df['roi'] != 'CSF'][['roi', 'type', 'level', 'volume']].groupby('roi', as_index=False).sum()
all_regions_df
```

```
[7]:
```

	roi	type	level	volume
0	ACR_L	3	10	24800
1	ACR_R	3	10	26248
2	AGWM_L	1	5	2919
3	AGWM_R	1	5	3266
4	AG_L	2	9	13170
..	...	...	...	...
420	subcallosal_ACC_R	3	10	1095
421	subgenualWM_ACC_L	1	5	218
422	subgenualWM_ACC_R	1	5	106
423	subgenual_ACC_L	3	10	3026
424	subgenual_ACC_R	3	10	2571

[425 rows x 4 columns]

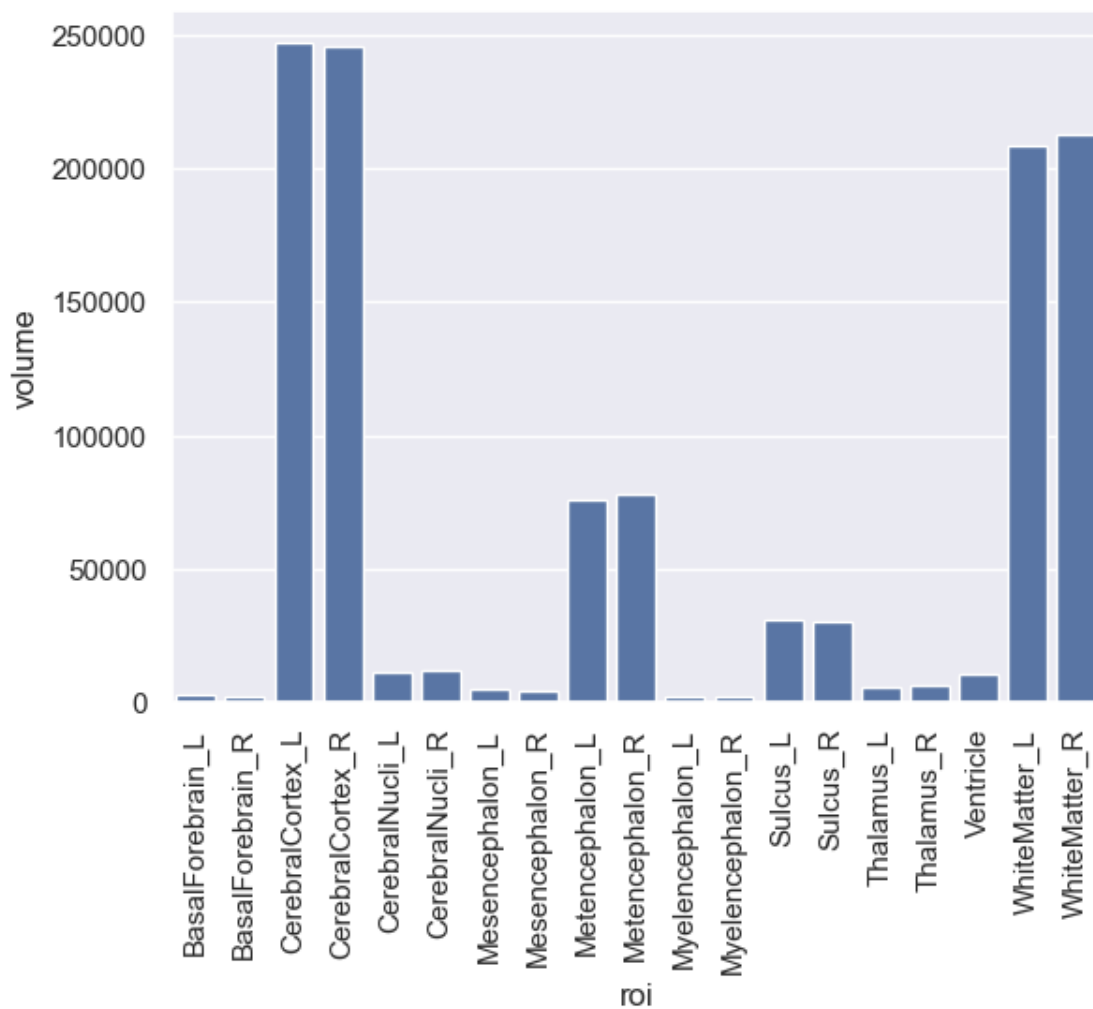
```
[8]: all_regions_t1l2_df = kirby906a_ax_df.loc[
    (kirby906a_ax_df['roi'] != 'CSF') & (kirby906a_ax_df['type'] == 1) &
    (kirby906a_ax_df['level'] == 2)
][['roi', 'type', 'level', 'volume']].groupby('roi', as_index=False).sum()
```

```
all_regions_t1l2_df.head()
```

```
[8]:
```

	roi	type	level	volume
0	BasalForebrain_L	1	2	2686
1	BasalForebrain_R	1	2	2609
2	CerebralCortex_L	1	2	246947
3	CerebralCortex_R	1	2	245855
4	CerebralNucli_L	1	2	11685

```
[9]: g = sns.barplot(x='roi', y='volume', data = all_regions_t1l2_df)
plt.xticks(rotation=90)
plt.show()
```



```
[10]: all_regions_t1l3_df = kirby906a_ax_df.loc[
```

```
(kirby906a_ax_df['roi'] != 'CSF') & (kirby906a_ax_df['type'] == 1) &
↳(kirby906a_ax_df['level'] == 3)
][['roi', 'type', 'level', 'volume']].groupby('roi', as_index=False).sum()
all_regions_t1l3_df.head(), all_regions_t1l3_df.tail()
```

```
[10]: (
      roi  type  level  volume
0   AnteriorWM_L    1     3   72167
1   AnteriorWM_R    1     3   76270
2  BasalForebrain_L    1     3    2686
3  BasalForebrain_R    1     3    2609
4    BasalGang_L    1     3   10250,
      roi  type  level  volume
49  Temporal_R    1     3   55776
50  Thalamus_L    1     3    6116
51  Thalamus_R    1     3    6446
52  midbrain_L    1     3    4842
53  midbrain_R    1     3    4722)
```

```
[11]: # 53 regions makes this chart hard to read,
# so I asked GPT how to display labels for only the 20 largest ROIs, and it
↳produced this cell
top20_idx = all_regions_t1l3_df['volume'].nlargest(20).index

def autopct_top20(pct):
    autopct_top20.i += 1
    idx = all_regions_t1l3_df.index[autopct_top20.i]
    return f'{pct:.1f}%' if idx in top20_idx else ''

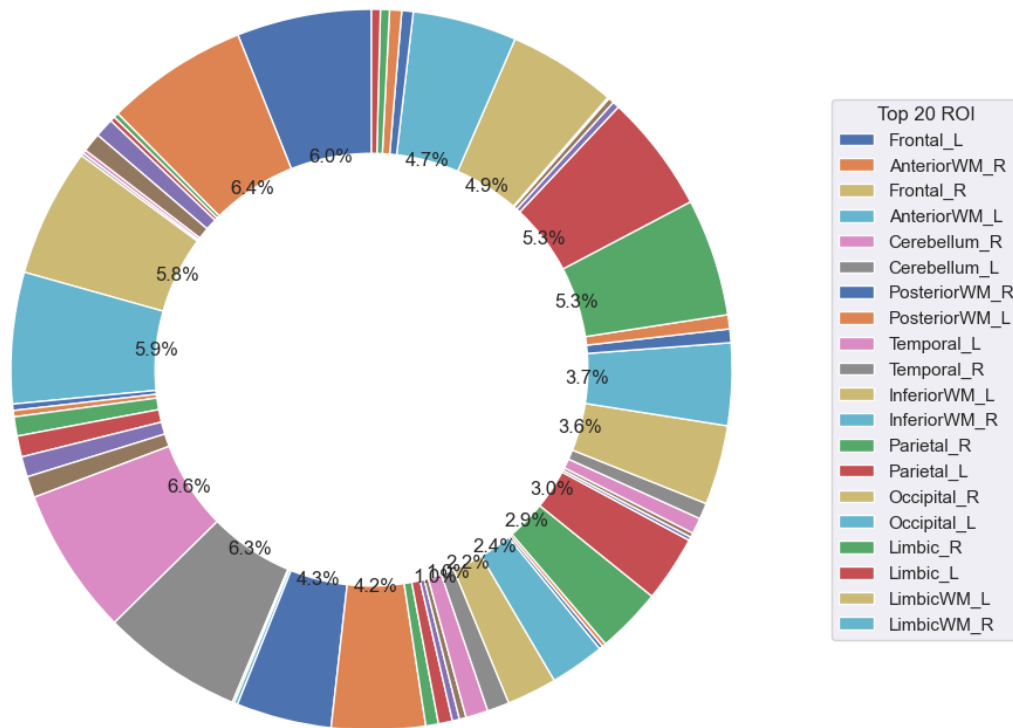
autopct_top20.i = -1
fig, ax = plt.subplots(figsize=(10, 10))

wedges, _, _ = ax.pie(
    all_regions_t1l3_df['volume'],
    autopct=autopct_top20,
    startangle=90,
    wedgeprops={'width': 0.4}
)

ax.legend(
    [wedges[i] for i in range(len(wedges)) if all_regions_t1l3_df.index[i] in
↳top20_idx],
    all_regions_t1l3_df.loc[top20_idx, 'roi'],
    title='Top 20 ROI',
    loc='center left',
    bbox_to_anchor=(1, 0.5)
)
```

```
ax.set_title('Type 1 Level 3 Volume by ROI')
plt.show()
```

Type 1 Level 3 Volume by ROI



```
[12]: class_interests_df = pd.read_csv("classInterests.txt", sep='\t')
class_interests_df.head()
```

```
[12]:
```

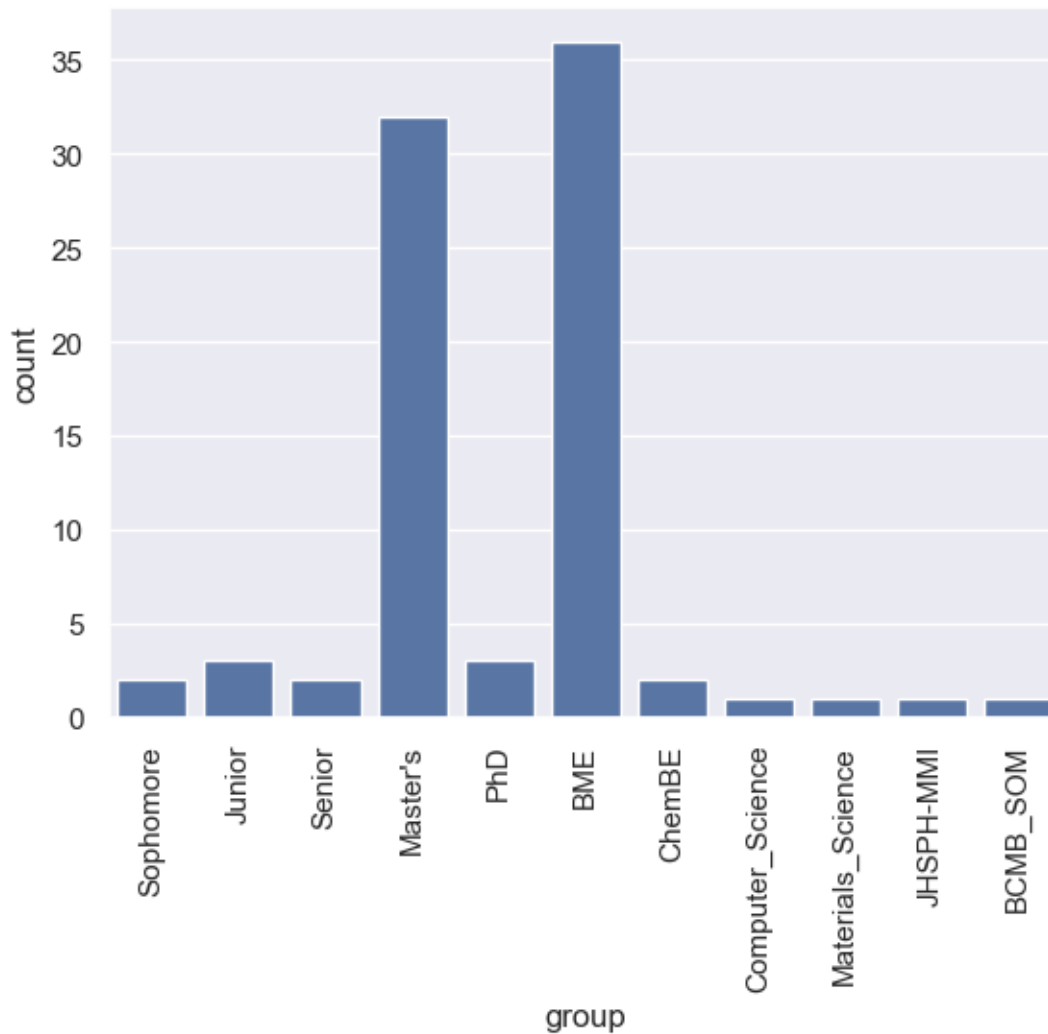
	Year	Program
0	Master's	BME
1	Master's	BME
2	Senior	BME
3	PhD	BME
4	PhD	Computer_Science

```
[13]: class_interests_counts_df = pd.concat([
    class_interests_df['Year'].value_counts(),
    class_interests_df['Program'].value_counts()
]).groupby(level=0).sum().reset_index()
class_interests_counts_df.columns = ['group', 'count']
```

```

g = sns.barplot(x='group', y='count', data=class_interests_counts_df,
               order=['Sophomore', 'Junior', 'Senior', "Master's", 'PhD',
                     'BME', 'ChemBE', 'Computer_Science', 'Materials_Science', 'JHSPH-MMI',
                     'BCMB_SOM'])
plt.xticks(rotation=90)
plt.show()

```



```

[14]: gene_expression_df = pd.read_csv("GSE5859_exprs.csv")
      gene_expression_df

```

```

[14]:      Unnamed: 0  GSM25581.CEL.gz  GSM25681.CEL.gz  \
0      1007_s_at      6.333951      5.736190
1      1053_at      7.255622      7.399993

```

2	117_at	5.760106	4.825169
3	121_at	6.935451	7.025210
4	1255_g_at	3.597872	3.108801
...	...	...	...
8788	AFFX-r2-Ec-bioC-5_at	7.403732	8.173182
8789	AFFX-r2-Ec-bioD-3_at	9.499829	10.147961
8790	AFFX-r2-Ec-bioD-5_at	9.087917	9.356494
8791	AFFX-r2-P1-cre-3_at	10.499176	11.039103
8792	AFFX-r2-P1-cre-5_at	10.959053	11.559275

	GSM136524.CEL.gz	GSM136707.CEL.gz	GSM25553.CEL.gz	GSM136676.CEL.gz	\
0	6.156754	6.513010	6.061013	6.506493	
1	7.491967	7.006123	7.250995	7.082581	
2	5.039387	5.414160	5.205697	5.300078	
3	7.543667	7.959781	7.223323	8.006816	
4	3.128269	3.364947	3.125909	3.186834	
...	...	...	...	...	
8788	7.973033	6.993441	6.699585	6.926527	
8789	11.527503	10.814206	10.629492	10.702125	
8790	11.116875	10.056376	10.344678	10.186448	
8791	2.721370	2.734304	5.357892	2.798261	
8792	2.850216	3.278276	6.196068	2.942734	

	GSM136711.CEL.gz	GSM136542.CEL.gz	GSM136535.CEL.gz	...	\
0	6.863426	6.369645	6.646321	...	
1	6.842236	7.048487	7.042361	...	
2	5.099337	5.156459	5.736109	...	
3	8.102504	7.434617	7.532321	...	
4	3.311022	3.092042	3.167737	...	
...	...	...	...	...	
8788	7.637367	7.006133	7.495841	...	
8789	11.285036	11.055667	11.388749	...	
8790	10.663689	10.411306	10.761674	...	
8791	2.802282	2.674753	2.881834	...	
8792	3.268022	2.936459	2.955473	...	

	GSM48650.CEL.gz	GSM25687.CEL.gz	GSM25685.CEL.gz	GSM136549.CEL.gz	\
0	7.082636	6.315866	7.010165	6.460107	
1	6.599718	7.412731	7.274429	6.928642	
2	6.231457	5.240717	5.275062	5.759827	
3	8.215096	7.677593	7.923624	7.219401	
4	3.331424	3.043856	3.151900	3.132253	
...	...	...	...	...	
8788	6.542924	6.701961	5.589173	6.999664	
8789	11.160386	11.321130	10.944250	11.098619	
8790	10.669841	10.725765	9.989154	10.428633	
8791	2.774188	5.756867	4.570855	2.633058	

8792	3.102871	6.470545	5.468001	2.937269
	GSM25427.CEL.gz	GSM25525.CEL.gz	GSM25349.CEL.gz	GSM136727.CEL.gz \
0	6.122355	6.330314	6.627014	6.133068
1	7.168782	7.235648	6.939184	7.280781
2	5.409720	5.491938	5.113570	5.401876
3	7.432338	6.785174	7.833862	7.607461
4	3.250600	2.984125	3.152269	3.225123
...	...	...	...	...
8788	7.852193	7.728074	8.243135	6.597565
8789	10.065078	10.222762	10.265151	11.160541
8790	9.288700	9.378446	9.569521	10.421423
8791	10.544423	10.714675	10.901787	2.535893
8792	11.080477	11.207895	11.396429	2.890163
	GSM25626.CEL.gz	GSM136725.CEL.gz		
0	6.419444	6.488579		
1	6.351776	7.517410		
2	5.537605	5.247190		
3	7.302935	7.331864		
4	3.222731	3.175846		
...	...	...		
8788	8.454131	6.635873		
8789	10.333127	10.762506		
8790	9.766366	10.108590		
8791	10.903407	2.813128		
8792	11.563012	2.789050		

[8793 rows x 209 columns]

```
[15]: gene_expression_df = gene_expression_df.drop(columns='Unnamed: 0')
gene_expression_df_pt1 = gene_expression_df.sub(gene_expression_df.
↳ mean(axis=1), axis=0)
gene_expression_df_pt1
```

[15]:	GSM25581.CEL.gz	GSM25681.CEL.gz	GSM136524.CEL.gz	GSM136707.CEL.gz \
0	-0.049313	-0.647073	-0.226509	0.129747
1	0.163992	0.308363	0.400337	-0.085507
2	0.283074	-0.651863	-0.437645	-0.062873
3	-0.648591	-0.558832	-0.040375	0.375740
4	0.400185	-0.088886	-0.069418	0.167261
...	...	...	...	...
8788	0.272703	1.042153	0.842004	-0.137588
8789	-1.086185	-0.438053	0.941489	0.228192
8790	-0.894001	-0.625424	1.134957	0.074458
8791	4.832515	5.372443	-2.945291	-2.932357
8792	4.853525	5.453746	-3.255313	-2.827252

	GSM25553.CEL.gz	GSM136676.CEL.gz	GSM136711.CEL.gz	GSM136542.CEL.gz	\
0	-0.322250	0.123230	0.480163	-0.013618	
1	0.159365	-0.009050	-0.249394	-0.043143	
2	-0.271335	-0.176954	-0.377696	-0.320573	
3	-0.360718	0.422774	0.518462	-0.149425	
4	-0.071777	-0.010852	0.113335	-0.105644	
...	...	...	...	...	
8788	-0.431444	-0.204502	0.506338	-0.124896	
8789	0.043478	0.116111	0.699022	0.469653	
8790	0.362761	0.204530	0.681772	0.429388	
8791	-0.308769	-2.868399	-2.864378	-2.991908	
8792	0.090539	-3.162795	-2.837506	-3.169069	

	GSM136535.CEL.gz	GSM25399.CEL.gz	...	GSM48650.CEL.gz	\
0	0.263058	-0.572333	...	0.699373	
1	-0.049269	-0.425005	...	-0.491913	
2	0.259077	0.330464	...	0.754425	
3	-0.051720	0.127419	...	0.631054	
4	-0.029949	-0.228108	...	0.133737	
...	...	...	...	...	
8788	0.364811	3.426462	...	-0.588105	
8789	0.802735	1.449105	...	0.574372	
8790	0.779757	1.402459	...	0.687923	
8791	-2.784827	6.584329	...	-2.892473	
8792	-3.150056	6.611836	...	-3.002658	

	GSM25687.CEL.gz	GSM25685.CEL.gz	GSM136549.CEL.gz	GSM25427.CEL.gz	\
0	-0.067397	0.626902	0.076844	-0.260908	
1	0.321100	0.182799	-0.162988	0.077152	
2	-0.236315	-0.201970	0.282794	-0.067313	
3	0.093552	0.339583	-0.364641	-0.151704	
4	-0.153830	-0.045787	-0.065434	0.052913	
...	...	...	...	...	
8788	-0.429068	-1.541857	-0.131365	0.721164	
8789	0.735116	0.358236	0.512605	-0.520936	
8790	0.743847	0.007236	0.446716	-0.693218	
8791	0.090207	-1.095805	-3.033603	4.877762	
8792	0.365016	-0.637528	-3.168259	4.974948	

	GSM25525.CEL.gz	GSM25349.CEL.gz	GSM136727.CEL.gz	GSM25626.CEL.gz	\
0	-0.052949	0.243751	-0.250195	0.036180	
1	0.144017	-0.152446	0.189150	-0.739855	
2	0.014905	-0.363462	-0.075156	0.060573	
3	-0.798867	0.249820	0.023419	-0.281107	
4	-0.213562	-0.045417	0.027436	0.025044	
...	...	...	...	...	

8788	0.597045	1.112106	-0.533464	1.323102
8789	-0.363252	-0.320863	0.574527	-0.252887
8790	-0.603472	-0.412396	0.439506	-0.215551
8791	5.048014	5.235127	-3.130768	5.236746
8792	5.102366	5.290901	-3.215366	5.457483

	GSM136725.CEL.gz
0	0.105316
1	0.425779
2	-0.229842
3	-0.252178
4	-0.021841
...	...
8788	-0.495156
8789	0.176492
8790	0.126673
8791	-2.853533
8792	-3.316479

[8793 rows x 208 columns]

```
[16]: gene_expression_df_pt2 = gene_expression_df_pt1.sub(gene_expression_df_pt1.
↳mean(axis=0), axis=1)
gene_expression_df_pt2
```

[16]:	GSM25581.CEL.gz	GSM25681.CEL.gz	GSM136524.CEL.gz	GSM136707.CEL.gz	\
0	-0.024490	-0.640032	-0.223988	0.114936	
1	0.188815	0.315404	0.402858	-0.100318	
2	0.307897	-0.644822	-0.435124	-0.077684	
3	-0.623768	-0.551790	-0.037853	0.360928	
4	0.425008	-0.081844	-0.066896	0.152450	
...	...	...	...	...	
8788	0.297526	1.049194	0.844525	-0.152399	
8789	-1.061362	-0.431011	0.944010	0.213381	
8790	-0.869178	-0.618382	1.137478	0.059647	
8791	4.857338	5.379484	-2.942770	-2.947168	
8792	4.878348	5.460788	-3.252792	-2.842063	

	GSM25553.CEL.gz	GSM136676.CEL.gz	GSM136711.CEL.gz	GSM136542.CEL.gz	\
0	-0.428929	0.109485	0.457383	-0.017008	
1	0.052687	-0.022794	-0.272174	-0.046533	
2	-0.378014	-0.190699	-0.400475	-0.323963	
3	-0.467397	0.409030	0.495682	-0.152814	
4	-0.178456	-0.024597	0.090556	-0.109034	
...	...	...	...	...	
8788	-0.538123	-0.218246	0.483558	-0.128286	
8789	-0.063201	0.102366	0.676242	0.466263	

8790	0.256082	0.190786	0.658992	0.425998
8791	-0.415447	-2.882144	-2.887158	-2.995298
8792	-0.016139	-3.176539	-2.860286	-3.172459

	GSM136535.CEL.gz	GSM25399.CEL.gz	...	GSM48650.CEL.gz	\
0	0.250137	-0.462337	...	0.678049	
1	-0.062190	-0.315009	...	-0.513237	
2	0.246157	0.440460	...	0.733100	
3	-0.064641	0.237415	...	0.609730	
4	-0.042870	-0.118111	...	0.112413	
...	...	...	...	...	
8788	0.351891	3.536458	...	-0.609429	
8789	0.789815	1.559101	...	0.553047	
8790	0.766836	1.512455	...	0.666599	
8791	-2.797747	6.694326	...	-2.913797	
8792	-3.162976	6.721832	...	-3.023982	

	GSM25687.CEL.gz	GSM25685.CEL.gz	GSM136549.CEL.gz	GSM25427.CEL.gz	\
0	-0.061456	0.665913	0.076758	-0.251423	
1	0.327042	0.221810	-0.163074	0.086637	
2	-0.230374	-0.162959	0.282708	-0.057828	
3	0.099493	0.378593	-0.364727	-0.142219	
4	-0.147889	-0.006776	-0.065520	0.062398	
...	...	...	...	...	
8788	-0.423127	-1.502846	-0.131451	0.730648	
8789	0.741058	0.397247	0.512519	-0.511451	
8790	0.749789	0.046247	0.446630	-0.683733	
8791	0.096148	-1.056795	-3.033689	4.887247	
8792	0.370958	-0.598517	-3.168345	4.984433	

	GSM25525.CEL.gz	GSM25349.CEL.gz	GSM136727.CEL.gz	GSM25626.CEL.gz	\
0	-0.031388	0.235241	-0.261128	0.041275	
1	0.165579	-0.160956	0.178217	-0.734760	
2	0.036467	-0.371972	-0.086090	0.065668	
3	-0.777306	0.241310	0.012486	-0.276012	
4	-0.192001	-0.053927	0.016503	0.030139	
...	...	...	...	...	
8788	0.618607	1.103596	-0.544398	1.328197	
8789	-0.341690	-0.329373	0.563594	-0.247792	
8790	-0.581910	-0.420906	0.428572	-0.210456	
8791	5.069575	5.226617	-3.141702	5.241841	
8792	5.123928	5.282391	-3.226299	5.462578	

	GSM136725.CEL.gz
0	0.090015
1	0.410478
2	-0.245143

3	-0.267479
4	-0.037142
...	...
8788	-0.510458
8789	0.161191
8790	0.111371
8791	-2.868834
8792	-3.331780

[8793 rows x 208 columns]

```
[17]: gene_expression_df_pt3 = gene_expression_df_pt2.div(gene_expression_df_pt2.
↳std(axis=0), axis=1)
gene_expression_df_pt3
```

```
[17]: GSM25581.CEL.gz GSM25681.CEL.gz GSM136524.CEL.gz GSM136707.CEL.gz \
0 -0.062810 -1.694428 -0.797346 0.362538
1 0.484259 0.835005 1.434085 -0.316432
2 0.789673 -1.707110 -1.548943 -0.245036
3 -1.599798 -1.460817 -0.134750 1.138467
4 1.090032 -0.216675 -0.238137 0.480868
...
8788 0.763075 2.777651 3.006323 -0.480709
8789 -2.722110 -1.141065 3.360467 0.673063
8790 -2.229208 -1.637114 4.049170 0.188144
8791 12.457774 14.241719 -10.475605 -9.296175
8792 12.511658 14.456963 -11.579215 -8.964646

GSM25553.CEL.gz GSM136676.CEL.gz GSM136711.CEL.gz GSM136542.CEL.gz \
0 -1.152413 0.366177 1.252187 -0.054154
1 0.141554 -0.076237 -0.745136 -0.148166
2 -1.015618 -0.637799 -1.096388 -1.031533
3 -1.255766 1.368016 1.357039 -0.486577
4 -0.479461 -0.082265 0.247916 -0.347177
...
8788 -1.445787 -0.729933 1.323845 -0.408476
8789 -0.169803 0.342368 1.851361 1.484632
8790 0.688022 0.638090 1.804134 1.356425
8791 -1.116192 -9.639443 -7.904224 -9.537356
8792 -0.043361 -10.624061 -7.830657 -10.101457

GSM136535.CEL.gz GSM25399.CEL.gz ... GSM48650.CEL.gz \
0 1.008166 -0.730639 ... 1.446603
1 -0.250652 -0.497815 ... -1.094982
2 0.992122 0.696068 ... 1.564055
3 -0.260531 0.375192 ... 1.300847
4 -0.172784 -0.186654 ... 0.239831
```

...	...	...	...	...
8788	1.418280	5.588729	...	-1.300205
8789	3.183311	2.463877	...	1.179916
8790	3.090696	2.390161	...	1.422176
8791	-11.276184	10.579166	...	-6.216530
8792	-12.748221	10.622634	...	-6.451607

	GSM25687.CEL.gz	GSM25685.CEL.gz	GSM136549.CEL.gz	GSM25427.CEL.gz	\
0	-0.185273	1.615970	0.245537	-0.568262	
1	0.985940	0.538265	-0.521649	0.195814	
2	-0.694512	-0.395453	0.904341	-0.130702	
3	0.299944	0.918732	-1.166704	-0.321440	
4	-0.445844	-0.016443	-0.209588	0.141031	
...	...	...	...	...	
8788	-1.275609	-3.646955	-0.420492	1.651396	
8789	2.234082	0.963998	1.639471	-1.155970	
8790	2.260404	0.112226	1.428700	-1.545359	
8791	0.289860	-2.564523	-9.704306	11.046052	
8792	1.118334	-1.452421	-10.135050	11.265710	

	GSM25525.CEL.gz	GSM25349.CEL.gz	GSM136727.CEL.gz	GSM25626.CEL.gz	\
0	-0.069236	0.617954	-0.598607	0.088393	
1	0.365237	-0.422815	0.408542	-1.573530	
2	0.080440	-0.977132	-0.197351	0.140632	
3	-1.714599	0.633897	0.028623	-0.591095	
4	-0.423519	-0.141661	0.037831	0.064544	
...	...	...	...	...	
8788	1.364537	2.899034	-1.247970	2.844410	
8789	-0.753708	-0.865229	1.291974	-0.530662	
8790	-1.283592	-1.105678	0.982453	-0.450704	
8791	11.182587	13.729787	-7.201991	11.225705	
8792	11.302478	13.876300	-7.395920	11.698425	

	GSM136725.CEL.gz
0	0.257410
1	1.173822
2	-0.701024
3	-0.764897
4	-0.106213
...	...
8788	-1.459729
8789	0.460950
8790	0.318483
8791	-8.203855
8792	-9.527717

[8793 rows x 208 columns]

```
[18]: healthcare_df = pd.read_csv("healthcare-spending.csv", skiprows=2,
    ↪skipfooter=16)
healthcare_df = healthcare_df[healthcare_df['Location'] != 'United States'].
    ↪reset_index(drop=True)
healthcare_df
```

C:\Users\Shane\AppData\Local\Temp\ipykernel_15308\3078209122.py:1:
 ParserWarning: Falling back to the 'python' engine because the 'c' engine does
 not support skipfooter; you can avoid this warning by specifying
 engine='python'.

```
healthcare_df = pd.read_csv("healthcare-spending.csv", skiprows=2,
skipfooter=16)
```

```
[18]:
```

	Location	1991__Total Health Spending \
0	Alabama	10393
1	Alaska	1458
2	Arizona	9269
3	Arkansas	5632
4	California	81438
5	Colorado	8460
6	Connecticut	10950
7	Delaware	1938
8	District of Columbia	2800
9	Florida	39430
10	Georgia	16612
11	Hawaii	2959
12	Idaho	2213
13	Illinois	31577
14	Indiana	14272
15	Iowa	7248
16	Kansas	6529
17	Kentucky	8877
18	Louisiana	11259
19	Maine	3152
20	Maryland	13560
21	Massachusetts	20015
22	Michigan	25038
23	Minnesota	11981
24	Mississippi	5650
25	Missouri	13012
26	Montana	1907
27	Nebraska	3790
28	Nevada	3086
29	New Hampshire	2727
30	New Jersey	23100
31	New Mexico	3483
32	New York	58080

33	North Carolina	15352
34	North Dakota	1709
35	Ohio	29348
36	Oklahoma	7512
37	Oregon	6760
38	Pennsylvania	36171
39	Rhode Island	2967
40	South Carolina	8075
41	South Dakota	1707
42	Tennessee	12692
43	Texas	40967
44	Utah	3522
45	Vermont	1330
46	Virginia	14829
47	Washington	12674
48	West Virginia	4672
49	Wisconsin	12694
50	Wyoming	1023

	1992__Total Health Spending	1993__Total Health Spending \
0	11284	12028
1	1558	1661
2	9815	10655
3	6022	6397
4	87949	91963
5	9215	9803
6	11635	12081
7	2111	2285
8	3098	3240
9	43041	46123
10	18207	19722
11	3163	3439
12	2415	2608
13	34119	36158
14	15552	16754
15	7704	8132
16	7001	7344
17	9697	10280
18	12286	13170
19	3326	3567
20	14531	15528
21	21268	22863
22	26437	28022
23	12598	13458
24	6060	6570
25	14242	15186
26	2071	2187

27	4113	4352
28	3464	3696
29	2997	3131
30	25192	26597
31	3823	4070
32	62675	67025
33	16855	18457
34	1845	1941
35	31928	33903
36	8178	8559
37	7458	7960
38	39165	41395
39	3205	3396
40	8830	9536
41	1825	1924
42	13979	15165
43	44949	48177
44	3795	4143
45	1421	1522
46	15599	16634
47	13859	14523
48	5159	5550
49	13669	14636
50	1067	1171

	1994__Total Health Spending	1995__Total Health Spending \
0	12742	13590
1	1728	1879
2	11364	12042
3	6810	7343
4	94245	96870
5	10382	11153
6	12772	13649
7	2489	2655
8	3255	3285
9	48886	51568
10	20784	22276
11	3710	3972
12	2762	3005
13	38139	40104
14	17425	18302
15	8619	9085
16	7829	8367
17	10768	11618
18	13926	14578
19	3719	4052
20	16540	17273

21	23792	25226
22	29729	31842
23	14555	15634
24	7077	7761
25	16307	17099
26	2286	2498
27	4630	4901
28	4102	4404
29	3299	3642
30	28432	30305
31	4322	4713
32	70819	74482
33	20015	21733
34	2076	2261
35	35399	37365
36	9187	9910
37	8381	9022
38	43035	45388
39	3444	3812
40	10213	10866
41	2089	2269
42	16005	17405
43	50473	54434
44	4359	4700
45	1625	1793
46	17637	18648
47	15303	16515
48	5891	6304
49	15532	16601
50	1265	1378

	1996__Total Health Spending	1997__Total Health Spending \
0	14450	15462
1	2076	2240
2	12850	13418
3	7817	8393
4	100215	103681
5	11863	12572
6	14139	14948
7	2772	3026
8	3362	3374
9	55090	58706
10	23622	24976
11	4246	4263
12	3260	3509
13	42069	44152
14	18747	20220

15	9667	10183
16	8867	9428
17	12473	13533
18	15125	15920
19	4395	4752
20	18231	19093
21	26553	28098
22	33797	35562
23	16559	17639
24	8339	8947
25	18154	19263
26	2577	2809
27	5325	5621
28	4769	5211
29	3963	4195
30	31420	33093
31	5090	5436
32	78348	81674
33	23678	25569
34	2365	2449
35	39749	40948
36	10590	11013
37	9530	10217
38	46598	49227
39	3950	4186
40	11487	12677
41	2435	2595
42	18603	19817
43	58352	63245
44	5153	5700
45	1894	2026
46	19958	21258
47	17158	17978
48	6632	7031
49	17702	18628
50	1476	1568

	1998__Total Health Spending	1999__Total Health Spending	...	\
0	15860	16451	...	
1	2386	2569	...	
2	14465	15550	...	
3	8814	9407	...	
4	111224	116036	...	
5	13790	14764	...	
6	15944	16785	...	
7	3207	3539	...	
8	3461	3578	...	

9	61558	65516	...
10	26480	27814	...
11	4446	4570	...
12	3781	4052	...
13	46490	48640	...
14	21574	22993	...
15	11057	11740	...
16	10071	10723	...
17	14337	15409	...
18	16532	17437	...
19	5151	5604	...
20	19538	20986	...
21	30165	31012	...
22	36417	38058	...
23	19292	20475	...
24	9334	9714	...
25	20574	21727	...
26	2989	3224	...
27	6026	6461	...
28	5694	6249	...
29	4545	4907	...
30	35003	36634	...
31	5689	6003	...
32	86599	91675	...
33	26873	29178	...
34	2605	2683	...
35	42515	44823	...
36	11582	12266	...
37	11369	12100	...
38	51278	54667	...
39	4460	4696	...
40	13422	14444	...
41	2728	2967	...
42	20700	21703	...
43	67106	71862	...
44	5989	6159	...
45	2172	2388	...
46	22391	23707	...
47	19746	21504	...
48	7343	7671	...
49	19640	21373	...
50	1690	1783	...
	2005__Total Health Spending	2006__Total Health Spending	\
0	25338	26638	
1	4765	5048	
2	28190	30766	

3	14611	15431
4	182958	194413
5	22867	24849
6	24538	25997
7	5899	6285
8	4971	5138
9	106293	114162
10	43655	47131
11	6967	7275
12	6854	7426
13	71747	76296
14	35391	37908
15	17241	18411
16	15885	16689
17	23709	25217
18	24866	25573
19	9233	9760
20	33911	36747
21	47920	51637
22	55564	59457
23	32417	34407
24	15653	16826
25	32853	33959
26	5107	5472
27	10491	11300
28	11995	12991
29	8143	8809
30	55670	58264
31	9702	10316
32	135771	143635
33	48395	52192
34	4009	4269
35	70218	73333
36	19040	20416
37	19495	21077
38	82123	86069
39	7497	7815
40	23070	24640
41	4564	4900
42	34589	36229
43	115264	123253
44	10812	11570
45	4080	4319
46	39148	41911
47	34101	36500
48	11643	12258
49	34078	36475

50	2922	3212
	2007__Total Health Spending	2008__Total Health Spending \
0	27700	28765
1	5426	5807
2	33366	35547
3	16426	17246
4	209397	221013
5	26525	27797
6	27488	29141
7	6735	7191
8	5492	5779
9	120960	126977
10	49827	50930
11	8023	8503
12	7990	8377
13	81934	84809
14	40055	41200
15	19327	20184
16	17684	18502
17	26396	27282
18	27532	29353
19	10243	10725
20	39047	40931
21	55554	58260
22	62551	64824
23	36648	37934
24	18083	18931
25	37229	39643
26	5868	6223
27	11883	12540
28	14123	14907
29	9442	9967
30	61598	63759
31	11089	11896
32	150601	155573
33	54832	57844
34	4576	4892
35	76591	78831
36	21924	23357
37	22264	23390
38	91040	94455
39	8196	8523
40	26169	27894
41	5208	5556
42	38407	39387
43	130965	139968

44	12466	13287
45	4548	4758
46	44769	47819
47	39595	42569
48	12996	13482
49	38529	40442
50	3501	3719

	2009__Total Health Spending	2010__Total Health Spending \
0	30095	30728
1	6112	6519
2	37258	38620
3	18071	18735
4	229541	241916
5	29246	30187
6	31132	31727
7	7495	7938
8	6182	6582
9	133067	137609
10	53035	53950
11	8810	8860
12	8861	9424
13	88515	93140
14	43867	45352
15	21067	21895
16	19161	19605
17	28917	29995
18	31253	32848
19	11114	11338
20	43019	44847
21	61376	63153
22	67489	70336
23	39720	41333
24	19572	19729
25	41141	42659
26	6594	6968
27	13038	13769
28	15302	15652
29	10705	11149
30	67653	68477
31	12657	13294
32	164928	170644
33	61730	63235
34	5266	5615
35	84413	86841
36	24179	24993
37	24697	25824

38	97553	103235
39	8843	9026
40	29204	30382
41	5919	6289
42	40982	42117
43	148903	155553
44	13893	14350
45	5068	5314
46	51141	53048
47	45589	47077
48	14360	14760
49	42586	43867
50	3903	4121

	2011__Total Health Spending	2012__Total Health Spending \
0	31398	32848
1	6928	7406
2	39295	40495
3	19356	20076
4	253844	266767
5	31372	32726
6	32129	33421
7	8365	8650
8	7000	7130
9	141462	147685
10	56053	59575
11	8993	9469
12	9718	10182
13	95542	98651
14	47476	51306
15	22733	23528
16	20522	21573
17	31206	31960
18	32765	33618
19	11720	11961
20	46381	47792
21	64915	67055
22	73143	75507
23	42618	43997
24	20351	21995
25	44726	46743
26	7285	7685
27	14213	14808
28	16198	16588
29	11558	11954
30	70263	73373
31	13644	14109

32	175996	177908
33	65703	68932
34	6003	6303
35	88339	93242
36	26205	27386
37	26964	27757
38	107458	110252
39	9243	9435
40	31339	32351
41	6649	6955
42	43134	45222
43	162296	170992
44	15042	15830
45	5562	5827
46	55772	58535
47	48567	51443
48	15336	16270
49	45511	46158
50	4289	4518

	2013__Total Health Spending	2014__Total Health Spending
0	33788	35263
1	7684	8151
2	41481	43356
3	20500	21980
4	278168	291989
5	34090	36398
6	34223	35413
7	9038	9587
8	7443	7871
9	150547	160624
10	62399	66447
11	9781	10338
12	10627	11315
13	101891	106306
14	52046	54741
15	24135	25487
16	21490	22183
17	33194	35323
18	34639	36324
19	12139	12684
20	48929	51330
21	68899	71274
22	76672	79874
23	45865	48377
24	22017	22879
25	47499	49137

26	8108	8409
27	15197	15823
28	17485	19020
29	12392	12742
30	75148	79066
31	14304	15027
32	183969	192809
33	69157	72160
34	6795	7289
35	95866	101013
36	28097	29575
37	29314	31920
38	113459	118419
39	9646	10071
40	33468	35299
41	7221	7616
42	46149	48249
43	176341	188559
44	16425	17597
45	6221	6389
46	60364	62847
47	53022	55819
48	16622	17491
49	47030	50109
50	4639	4856

[51 rows x 25 columns]

```
[19]: healthcare_df = healthcare_df.melt(id_vars='Location', var_name='Year',
    ↳value_name='Spending')
healthcare_df['Year'] = healthcare_df['Year'].str.extract(r'(\d{4})').
    ↳astype(int)
healthcare_df
```

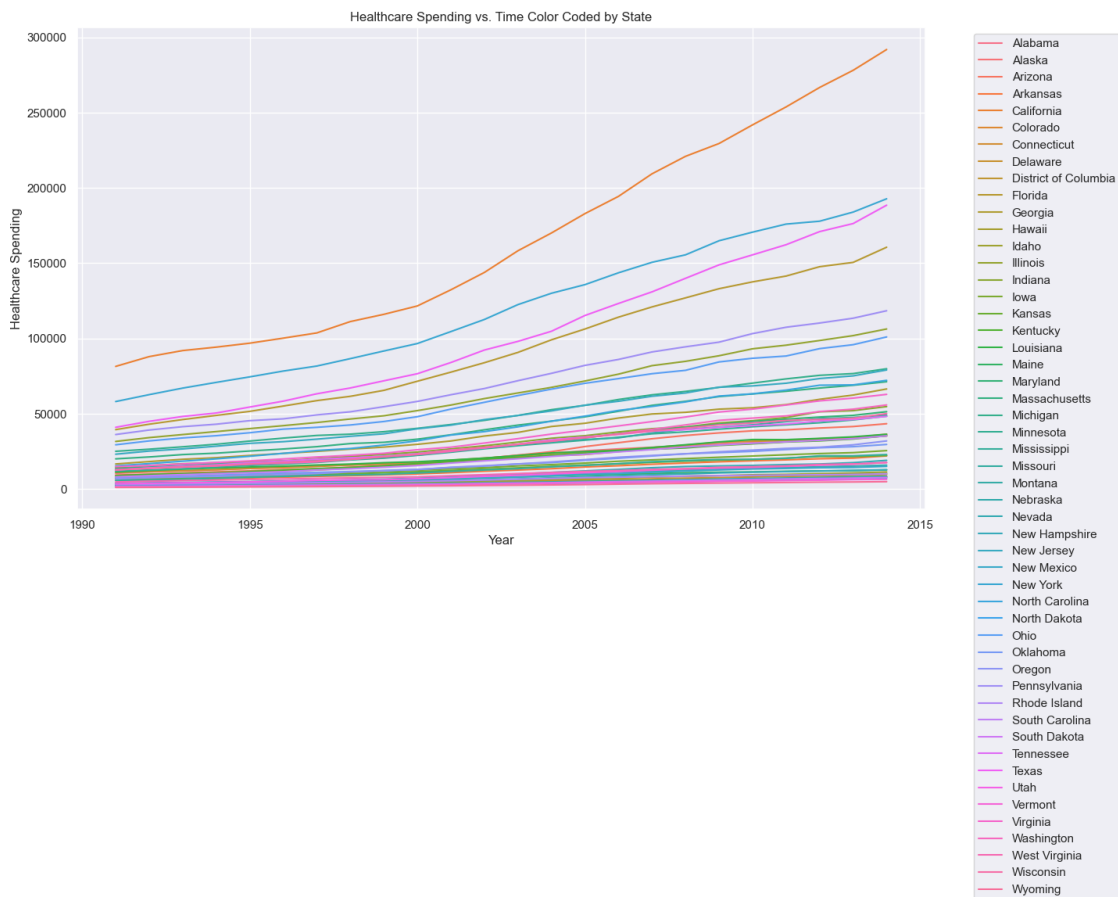
```
[19]:
```

	Location	Year	Spending
0	Alabama	1991	10393
1	Alaska	1991	1458
2	Arizona	1991	9269
3	Arkansas	1991	5632
4	California	1991	81438
...	...	...	...
1219	Virginia	2014	62847
1220	Washington	2014	55819
1221	West Virginia	2014	17491
1222	Wisconsin	2014	50109
1223	Wyoming	2014	4856

[1224 rows x 3 columns]

```
[24]: plt.figure(figsize=(14, 8))
sns.lineplot(data=healthcare_df, x='Year', y='Spending', hue='Location')

plt.title('Healthcare Spending vs. Time Color Coded by State')
plt.xlabel('Year')
plt.ylabel('Healthcare Spending')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```



```
[26]: healthcare_average_df = healthcare_df.groupby('Location')['Spending'].mean().
      ↪sort_values()
healthcare_average_df
```

```
[26]: Location
Wyoming                2655.041667
Vermont                3513.541667
North Dakota          3816.541667
```

Alaska	4141.541667
South Dakota	4153.750000
Montana	4614.750000
District of Columbia	4738.291667
Delaware	5167.416667
Idaho	6042.958333
Hawaii	6221.958333
Rhode Island	6348.458333
New Hampshire	7142.000000
Maine	7711.583333
New Mexico	8577.750000
Nebraska	9188.291667
Utah	9440.291667
Nevada	9979.833333
West Virginia	10376.416667
Arkansas	12823.625000
Mississippi	13691.666667
Kansas	13975.083333
Iowa	15377.583333
Oklahoma	17108.458333
Oregon	17199.458333
South Carolina	20271.583333
Colorado	20581.291667
Kentucky	20765.708333
Alabama	21976.708333
Connecticut	22112.625000
Louisiana	22625.291667
Arizona	24176.083333
Minnesota	28264.541667
Missouri	29513.041667
Tennessee	29561.041667
Wisconsin	29722.000000
Maryland	30165.916667
Washington	30840.666667
Indiana	31661.875000
Virginia	35011.333333
Georgia	38304.250000
North Carolina	41838.083333
Massachusetts	43122.541667
New Jersey	48840.666667
Michigan	50023.833333
Ohio	61323.333333
Illinois	64977.666667
Pennsylvania	72803.083333
Florida	93009.750000
Texas	102809.375000
New York	120398.416667

California 165824.708333
Name: Spending, dtype: float64

```
[27]: plt.figure(figsize=(14, 8))
plt.bar(healthcare_average_df.index, healthcare_average_df.values)
plt.xticks(rotation=90)
plt.xlabel('State')
plt.ylabel('Average Health Care Spending')
plt.title('Average Health Care Spending by State')

plt.tight_layout()
plt.show()
```

